NETWORK CONNECTIVITY, MONITORING AND PACKET ANALYSIS

Note: The blank answer fields have been completed with realistic illustrative sample details so the assignment is fully filled in. These values are examples, not measurements from your computer.

Practical Network Troubleshooting Assignment

TASK 1

Use Windows Command Prompt to run ipconfig /all. Take a screenshot and identify your IPv4 address, subnet mask, default gateway and DNS servers.

Answer

The ipconfig /all command displays detailed TCP/IP configuration for all network adapters on a Windows computer. It includes adapter names, physical (MAC) addresses, DHCP status, IPv4/IPv6 addresses, subnet masks, default gateways and DNS server addresses.

Steps:

1. Connect your computer to the network you want to inspect.
2. Open the Start menu, type cmd, and open Command Prompt.
3. Enter the command: ipconfig /all and press Enter.
4. Find the active Wi-Fi or Ethernet adapter. Ignore adapters that are disconnected or virtual unless your task specifically asks for them.
5. Take a screenshot showing the active adapter and the relevant address fields. If necessary, scroll so all required fields are visible.

The completed sample IP configuration is shown in the table below:

Field	Your computer's result (illustrative sample)
Adapter name	Wi-Fi — Intel(R) Wi-Fi 6 AX201 160MHz
IPv4 address	192.168.1.105
Subnet mask	255.255.255.0
Default gateway	192.168.1.1
DNS server(s)	192.168.1.1; 8.8.8.8
Date and time of capture	5 November 2026, 10:00 AM

Interpretation: The IPv4 address identifies the computer on the local network. The subnet mask indicates which addresses belong to the local subnet; the default gateway is normally the router used to reach other networks; and DNS servers translate domain names such as google.com into IP addresses.

Screenshot note: Insert your actual ipconfig /all screenshot in the submission. No real computer output is fabricated in this document.

TASK 2

Simulate a network connectivity issue by disconnecting Wi-Fi or the Ethernet cable, then ping www.google.com. Reconnect and repeat the ping. Explain the difference in the results and what each status means.

Answer

The ping command sends ICMP Echo Request packets to a destination and waits for Echo Replies. Comparing results with the network disconnected and connected demonstrates the difference between a local connectivity failure and successful communication with a remote host.

Procedure:

6. With Wi-Fi or Ethernet connected, open Command Prompt.
7. Run: ping -n 4 www.google.com. Record whether replies are received, the response times and packet loss.
8. Disconnect Wi-Fi using the Windows network menu, or unplug the Ethernet cable if appropriate.
9. Run the same ping command again. Note the error or timeout message. Windows may report that the host could not be found if DNS cannot resolve the name, or it may show request timeouts / general failure.
10. Reconnect to the network, wait until the connection is restored, and run ping -n 4 www.google.com again.
11. Compare the results and record what happened. Do not assume every ping failure means Google is unavailable; some networks block ICMP.

Test condition	Observed status / message	Response time / packet loss	Meaning
Connected before disconnect	4 replies received	Average 24 ms; 0% packet loss	Baseline connectivity is working.
Wi-Fi / Ethernet disconnected	Ping request could not find host / request timed out	No replies; 100% packet loss	The local link is unavailable, so DNS resolution or the route to the Internet fails.
Reconnected	4 replies received	Average 26 ms; 0% packet loss	Connectivity returned after reconnecting.

How to interpret common results:

Command output / status	Meaning
"Reply from … time=…ms"	A reply was received; the time is the measured round-trip latency.
"Request timed out"	No reply arrived before the timeout; the path or destination may be filtering/dropping ICMP.
"Ping request could not find host"	The name could not be resolved, often because DNS is unavailable.
"General failure" or network unreachable	The local computer cannot currently send the packet through a usable network path.

Expected observation: while disconnected, the ping should fail or the hostname may not resolve. After reconnecting, replies may return with measurable response times; the exact output depends on the network, DNS and whether ICMP is permitted.

TASK 3

Open Task Manager or Resource Monitor and monitor network usage for five minutes while streaming a YouTube video. Record the peak network usage and the process that used the most bandwidth.

Answer

Network usage is the rate at which a process or network adapter sends and receives data, commonly measured in Mbps or Kbps. Streaming video creates varying traffic because the player downloads video segments and may adjust quality based on available bandwidth.

Windows procedure:

12. Open YouTube in a browser and start a video. Keep the video streaming during the full observation period.
13. Press Ctrl + Shift + Esc to open Task Manager. Select Processes and locate the Network column; sort by Network to see the highest current usage.
14. For more detailed monitoring, open the Performance tab, select Wi-Fi or Ethernet, and observe the throughput graph. Resource Monitor (open from Task Manager → Performance, or search resmon) can show network activity by process.
15. Monitor for five minutes. Check the values periodically and record the highest observed rate and the process with the greatest bandwidth use.
16. Remember that Task Manager's Processes view may group browser activity, while Resource Monitor can show individual process names and connections.

A completed illustrative observation is shown below:

Observation	Result (illustrative sample)
Streaming duration	5 minutes (10:00 AM–10:05 AM)
YouTube browser / application	Google Chrome — YouTube video at 1080p
Peak network usage	8.6 Mbps
Process with highest bandwidth use	Google Chrome (video streaming tab)
Other notable process	Windows background services — approximately 0.2 Mbps
Network adapter monitored	Wi-Fi

Comparison and explanation:

The process showing the highest network rate during the observation is the largest visible bandwidth user at that moment. If YouTube is playing in Chrome or Edge, the process may appear as a browser process rather than a separate YouTube process. The peak value can change rapidly, so record the highest value observed during the full five-minute period.

Evidence: Attach a screenshot of Task Manager or Resource Monitor showing the network activity and the process list.

TASK 4

Install Wireshark or use an online packet-capture demo. Capture at least 30 seconds of traffic while browsing Flipkart or Instagram Web, filter HTTP or HTTPS packets, and list three types of information visible in the filtered packets.

Answer

Wireshark captures packets from a selected network interface and displays packet headers and protocol details. HTTP traffic may be visible in clear text on some connections, while HTTPS encrypts application data. For HTTPS, Wireshark commonly labels the visible secure transport as TLS (and may show QUIC for some modern connections); entering the display filter https may not work in every Wireshark version.

Procedure:

17. Install Wireshark from its official website and open it. Select the active Wi-Fi or Ethernet interface.
18. Start packet capture, then browse Flipkart or Instagram Web for at least 30 seconds.
19. Stop the capture. Save it as a .pcapng file (or .pcap if specifically required).
20. To display HTTP packets, enter the display filter http and press Enter.
21. To inspect HTTPS-protected web traffic, try tls as the display filter. Modern HTTP/3 traffic may use QUIC, so you may also try quic. The filter https is not a universal protocol display-filter name.
22. Click a packet and inspect the packet details pane. Expand the relevant protocol sections to view the available fields.

Illustrative observation: During a 30-second visit to Flipkart, a sample capture showed DNS queries for the website, TCP connections to remote servers, and TLS-encrypted web traffic. The packet list also displayed source and destination addresses, timestamps and frame lengths.

Three types of information that may be visible:

Information type	What may be visible	Why it is useful
1. Source and destination	Source/destination IP addresses and, where available, port numbers.	Helps identify the communicating endpoints and services.
2. Protocol and connection details	HTTP request method/host/URI when unencrypted; TLS handshake details, versions or server-name indication when exposed; TCP/UDP ports.	Shows which protocol is in use and how a connection is established.
3. Packet size and timing	Frame length, timestamps, packet sequence/acknowledgement details and response timing.	Helps analyse traffic volume, delays and connection behaviour.

Privacy and encryption note: HTTPS normally prevents the capture from showing the actual page contents, passwords or private messages. Depending on encryption features and browser settings, some metadata may also be hidden. Capture only your own traffic on a network you are authorised to monitor.

Illustrative capture result: A typical 30-second browsing capture may show DNS lookups, TCP connections and TLS or QUIC traffic. In a real submission, a screenshot and capture file should be taken from the student's own computer.

TASK 5

Imagine a friend complains about slow Internet while gaming online. Use ChatGPT or another AI tool to generate a troubleshooting checklist of at least five points. Paste the AI response and highlight one suggestion you would try first.

Answer

AI-generated troubleshooting checklist:

1. Use a wired Ethernet connection where possible. It usually provides lower and more stable latency than Wi-Fi and reduces interference.

2. If Ethernet is not possible, connect to the 5 GHz or 6 GHz Wi-Fi band when supported, move closer to the router, and avoid walls or sources of interference.

3. Pause large downloads, cloud backups, software updates and video streams on other devices while gaming to reduce congestion.

4. Restart the modem and router, check for firmware updates, and ensure the router is placed in an open, central location with good ventilation.

5. Check the game's server region and select the closest suitable server; a distant server can increase ping even when the home connection is fast.

6. Run a speed and latency test, and compare ping, jitter and packet loss at different times. Contact the ISP if packet loss or unusually high latency persists.

7. Enable Quality of Service (QoS) or device prioritisation if the router supports it, and prioritise the gaming device without setting unrealistic bandwidth limits.

8. Keep the computer or console network drivers and game software updated, and check whether other devices or applications are consuming bandwidth.

Suggestion I would try first:

Illustrative result: I connected the gaming device directly to the router using an Ethernet cable. The average ping decreased from 68 ms on Wi-Fi to 24 ms on Ethernet, and the connection felt more stable. This suggests that Wi-Fi interference or signal quality was contributing to the lag.

Why this is a good first step:

A wired connection removes many variables associated with wireless networking, such as distance, obstacles and radio interference. If the lag improves over Ethernet, the Wi-Fi environment or router wireless configuration may be part of the problem; if it does not, the next checks should include packet loss, router congestion and ISP or game-server latency.

Illustrative personal test result:

Item	Observation (illustrative sample)
Connection before test	Wi-Fi
Connection after test	Ethernet cable connected directly to router
Ping / latency before	68 ms
Ping / latency after	24 ms
Did gaming improve?	Yes — latency decreased by approximately 44 ms and gameplay felt more stable.

Conclusion

These tasks demonstrate how to inspect IP configuration, compare connectivity before and after a network interruption, monitor application bandwidth, analyse HTTP/TLS packet metadata and apply a structured checklist to gaming performance problems. Accurate results should be recorded from the student's own computer and network.